Chapter 9. Analysis of Variance

One-way and two-way layouts

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9.1 What is analysis of variance?

Motivation. We often ask whether a characteristic differs across several conditions — strength across production methods, yield across temperatures, recovery time across drugs. Analysis of variance answers this by splitting the total variation of the data into parts due to each factor and a part due to error.

Such questions are studied by planned experiments. Suppose tensile strengths of metal parts made by several methods are compared and one method looks best. The difference may be due to the method, or to other factors such as the skill of the workers. To separate them we repeat each method several times (several workers, or one worker repeatedly). The observed variation then has a part due to differences between methods and a part due to other sources (worker skill, fluctuations of one worker). If the variation between methods is large relative to the variation from other sources, we conclude that the method affects strength.

Analysis of variance (ANOVA), devised by R. A. Fisher, is the most widely used tool in the **design of experiments** (DOE). It expresses the variation of the characteristic as a sum of squares, decomposes it into parts attributable to the factors of the experiment, and compares each factor's *mean square* (sum of squares divided by its degrees of freedom) with the mean square due to error. In the example, strength is the characteristic and the method is a factor; the total sum of squares splits as

$$\text{total sum of squares} = \text{sum of squares due to the factor} + \text{sum of squares due to error.}$$

The factor part has an identified cause (the method); the error part collects everything not yet explained. ANOVA generalizes the two-sample test of equal means (Chapter 6) to three or more means. This chapter treats *fixed* factors (levels chosen by the experimenter): one-way ANOVA (equal and unequal replication) and two-way ANOVA (without and with replication).

9.2 One-way analysis of variance

Motivation. The simplest design studies a single factor at $k \geq 3$ levels; the question is whether the k treatment means are all equal.

In a **one-way layout** a single factor A has k levels (treatments), and treatment i is replicated n_i times; the n_i need not be equal, and typically $k = 3$–5 and $n_i = 3$–10. The runs must be performed in random order: this is a **completely randomized design**. For example, with methods A, B, C and 5 runs each, number the $3 \times 5 = 15$ experimental units and run them in an order drawn by lot or random numbers. Statistically this amounts to independent random samples from k populations: from the $N = n_1 + \cdots + n_k$ units, n_1 are randomly chosen for treatment 1, n_2 of the rest for treatment 2, and so on. For $k = 2$ it is the comparison of two treatments based on independent samples.

9.2.1 Equal replication

With n replicates per treatment the data are y_{ij} ($i = 1, \ldots, k$ treatments, $j = 1, \ldots, n$), with treatment means and grand mean

$$\bar{y}_{i\cdot} = \frac{1}{n}\sum_{j=1}^{n} y_{ij}, \qquad \bar{y}_{\cdot\cdot} = \frac{1}{kn}\sum_{i=1}^{k}\sum_{j=1}^{n} y_{ij} \qquad (i = 1, \ldots, k).$$

The observations under treatment i are a random sample of size n from $N(\mu_i, \sigma^2)$: $Y_{ij} = \mu_i + e_{ij}$ with $e_{ij} \sim N(0, \sigma^2)$ independent. Let $\mu = \frac{1}{k}\sum \mu_i$ be the overall mean and $\alpha_i = \mu_i - \mu$ the ith **treatment effect**; by definition $\sum \alpha_i = 0$.

Definition (One-way ANOVA model (equal replication))

$$Y_{ij} = \mu + \alpha_i + e_{ij}, \qquad e_{ij} \sim N(0, \sigma^2) \text{ independent}, \qquad \sum_{i=1}^{k} \alpha_i = 0,$$

$i = 1, \dots, k$, $j = 1, \dots, n$. Equivalently:
(i) linearity $E(e_{ij}) = 0$, i.e. $E(Y_{ij}) = \mu + \alpha_i$;
(ii) equal variance $\mathrm{Var}(e_{ij}) = \sigma^2 > 0$;
(iii) independence of the e_{ij};
(iv) normality $e_{ij} \sim N(0, \sigma^2)$.

Testing whether the treatment effects differ is the same as testing whether the k population means differ. Decompose the total deviation:

$$y_{ij} - \bar{y}_{\cdot\cdot} = (y_{ij} - \bar{y}_{i\cdot}) + (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}). \tag{9.1}$$

Here $y_{ij} - \bar{y}_{i\cdot}$ is the **residual** (experimental uncertainty) and $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$ is the **deviation of the treatment effect**. Squaring and summing gives the following decomposition into the **total sum of squares** SST (total variation), the **treatment sum of squares** SS_{tr} (**between sum of squares**) and the **residual sum of squares** SSE (**within sum of squares**).

Proposition (Decomposition of the total sum of squares, one-way)

With $N = kn$ observations,

$$\underbrace{\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{\cdot\cdot})^2}_{\mathrm{SST},\ N-1} = \underbrace{n\sum_{i=1}^{k}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2}_{\mathrm{SS}_{tr},\ k-1} + \underbrace{\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{y}_{i\cdot})^2}_{\mathrm{SSE},\ N-k}.$$

Proof. [Supplement] Square (9.1) and sum over i, j. The cross term is

$$2\sum_{i=1}^{k}(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})\sum_{j=1}^{n}(y_{ij} - \bar{y}_{i\cdot}) = 0,$$

because the inner sum is $n\bar{y}_{i\cdot} - n\bar{y}_{i\cdot} = 0$. The middle square does not depend on j, so it contributes $n\sum_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$. As for degrees of freedom: SST has $N-1$ (the deviations sum to 0), SS_{tr} has $k-1$ (k treatment means minus one constraint), and SSE has $N-k$ ($n-1$ in each of k groups); indeed $N - 1 = (k-1) + (N-k)$. ■

A sum of squares divided by its degrees of freedom is a **mean square**:

$$\mathrm{MS}_{tr} = \frac{\mathrm{SS}_{tr}}{k-1}, \qquad \mathrm{MSE} = \frac{\mathrm{SSE}}{N-k}.$$

We test

$$H_0 : \alpha_1 = \alpha_2 = \cdots = \alpha_k = 0 \qquad \text{vs.} \qquad H_1 : \text{at least one } \alpha_i \neq 0,$$

which is the same as $H_0 : \mu_1 = \cdots = \mu_k$. If the treatments matter, SS_{tr} takes a large share of SST, so large values of $F = \mathrm{MS}_{tr}/\mathrm{MSE}$ are evidence against H_0.

Proposition (Expected mean squares, one-way)

$E(\mathrm{MSE}) = \sigma^2$ and $E(\mathrm{MS}_{tr}) = \sigma^2 + \dfrac{n}{k-1}\sum_{i=1}^{k}\alpha_i^2$. Under H_0, $F = \mathrm{MS}_{tr}/\mathrm{MSE} \sim F(k-1, N-k)$.

Proof. [Supplement]

Step 1. For each i, $Y_{i1}, \dots, Y_{in}$ is a random sample from $N(\mu_i, \sigma^2)$, so its sample variance has mean σ^2: $E\sum_j (Y_{ij} - \bar{Y}_{i\cdot})^2 = (n-1)\sigma^2$. Summing over i, $E(\mathrm{SSE}) = k(n-1)\sigma^2 = (N-k)\sigma^2$.

Step 2. The treatment means $\bar{Y}_{i\cdot} \sim N(\mu + \alpha_i, \sigma^2/n)$ are independent. For independent Z_i with means m_i and common variance v, $E\sum (Z_i - \bar{Z})^2 = (k-1)v + \sum (m_i - \bar{m})^2$. With $v = \sigma^2/n$, $m_i = \mu + \alpha_i$ and $\bar{m} = \mu$ (as $\sum \alpha_i = 0$),

$$E(\mathrm{SS}_{tr}) = n\Big\{(k-1)\frac{\sigma^2}{n} + \sum \alpha_i^2\Big\} = (k-1)\sigma^2 + n\sum \alpha_i^2.$$

Step 3. Distribution. $\mathrm{SSE}/\sigma^2 = \sum_i (n-1)S_i^2/\sigma^2$ is a sum of k independent $\chi^2(n-1)$ variables, hence $\chi^2(N-k)$. Each S_i^2 is independent of $\bar{Y}_{i\cdot}$ (normal sampling theory), and the groups are independent, so SSE is independent of $(\bar{Y}_{1\cdot}, \dots, \bar{Y}_{k\cdot})$, hence of SS_{tr}. Under H_0 the $\bar{Y}_{i\cdot}$ are i.i.d. $N(\mu, \sigma^2/n)$, so $\mathrm{SS}_{tr}/\sigma^2 = \sum (\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})^2/(\sigma^2/n) \sim \chi^2(k-1)$. Therefore $F = \dfrac{(\mathrm{SS}_{tr}/\sigma^2)/(k-1)}{(\mathrm{SSE}/\sigma^2)/(N-k)} \sim F(k-1, N-k)$.

■

Thus MSE always estimates σ^2, while MS_{tr} is inflated by treatment effects — which is why a large F indicates H_1.

Definition (Test of treatment effects (one-way, equal replication))

For $H_0 : \alpha_1 = \cdots = \alpha_k = 0$ use $F = \mathrm{MS}_{tr}/\mathrm{MSE}$. With observed value f:
(i) level-α critical region: $f \geq F_\alpha(k-1, N-k)$;
(ii) P-value: $P = P(F \geq f)$, $F \sim F(k-1, N-k)$.

Source	SS	d.f.	MS	F	P
Treatment	$\mathrm{SS}_{tr} = n\sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$	$k-1$	$\mathrm{MS}_{tr} = \frac{\mathrm{SS}_{tr}}{k-1}$	$f = \frac{\mathrm{MS}_{tr}}{\mathrm{MSE}}$	$P(F \geq f)$
Residual	$\mathrm{SSE} = \sum_i \sum_j (y_{ij} - \bar{y}_{i\cdot})^2$	$N-k$	$\mathrm{MSE} = \frac{\mathrm{SSE}}{N-k}$		
Total	$\mathrm{SST} = \sum_i \sum_j (y_{ij} - \bar{y}_{\cdot\cdot})^2$	$N-1$			

Example 9-1 (Concentration of a treatment solution)

To see whether the concentration of a treatment solution affects the tensile strength of a fabric, four concentrations (A_1: 3.0%, A_2: 3.5%, A_3: 4.0%, A_4: 4.5%) were each run 5 times, the 20 runs in random order. Build the ANOVA table and test at level 5%.

	A_1	A_2	A_3	A_4
	47	51	50	22
	58	62	38	23
	51	31	47	28
	61	46	27	42
	46	49	23	25
mean	52.6	47.8	37.0	28.0

grand mean 41.35

Solution. Model $Y_{ij} = \mu + \alpha_i + e_{ij}$ with $k = 4$, $n = 5$; $H_0 : \alpha_1 = \cdots = \alpha_4 = 0$.

$$\mathrm{SST} = (47 - 41.35)^2 + \cdots + (25 - 41.35)^2 = 3334.55,$$
$$\mathrm{SS}_{tr} = 5\{(52.6 - 41.35)^2 + \cdots + (28.0 - 41.35)^2\} = 1826.55,$$
$$\mathrm{SSE} = (47 - 52.6)^2 + \cdots + (25 - 28.0)^2 = 1508.00,$$

with d.f. 19, 3, 16. So $\mathrm{MS}_{tr} = 608.85$, $\mathrm{MSE} = 94.25$, $f = 6.46$, and $P = P(F(3,16) \geq 6.46) =$

0.0045 (table: between 0.001 and 0.01). Concentration has a significant effect on strength at level 5%.

Source	SS	d.f.	MS	F	P
Treatment	1826.55	3	608.85	6.46	0.0045
Residual	1508.00	16	94.25		
Total	3334.55	19			

9.2.2 Unequal replication

In practice the number of replicates often varies by treatment (e.g. more runs for a treatment of special interest). The analysis is almost the same. With n_i replicates of treatment i and $N = \sum n_i$,

$$\bar{y}_{i\cdot} = \frac{1}{n_i}\sum_{j=1}^{n_i} y_{ij}, \qquad \bar{y}_{\cdot\cdot} = \frac{1}{N}\sum_{i=1}^{k}\sum_{j=1}^{n_i} y_{ij}, \qquad \mu = \frac{1}{N}\sum_{i=1}^{k} n_i\mu_i,$$

and $\alpha_i = \mu_i - \mu$ now satisfies $\sum n_i\alpha_i = 0$.

Definition (One-way ANOVA model (unequal replication))

$Y_{ij} = \mu + \alpha_i + e_{ij}$, $e_{ij} \sim N(0,\sigma^2)$ independent, $\sum_{i=1}^{k} n_i\alpha_i = 0$ $(i = 1,\dots,k,\ j = 1,\dots,n_i)$.

Proposition (Decomposition and expected mean squares, unequal replication)

$$\sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{\cdot\cdot})^2 = \sum_{i=1}^{k} n_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 + \sum_{i=1}^{k}\sum_{j=1}^{n_i}(y_{ij} - \bar{y}_{i\cdot})^2,$$

with d.f. $N-1 = (k-1)+(N-k)$. Moreover $E(\mathrm{MSE}) = \sigma^2$, $E(\mathrm{MS}_{tr}) = \sigma^2 + \sum n_i\alpha_i^2/(k-1)$, and under $H_0 : \alpha_1 = \cdots = \alpha_k = 0$, $F = \mathrm{MS}_{tr}/\mathrm{MSE} \sim F(k-1, N-k)$.

Proof. [Supplement]

Step 1. The cross term is $2\sum_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})\sum_j(y_{ij} - \bar{y}_{i\cdot}) = 0$, exactly as before; the middle square summed over j gives $n_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$.

Step 2. $E(\mathrm{SSE}) = \sum(n_i - 1)\sigma^2 = (N-k)\sigma^2$ as before.

Step 3. Write $\mathrm{SS}_{tr} = \sum n_i\bar{Y}_{i\cdot}^2 - N\bar{Y}_{\cdot\cdot}^2$. Since $E\bar{Y}_{i\cdot}^2 = \sigma^2/n_i + \mu_i^2$ and $\bar{Y}_{\cdot\cdot} \sim N(\mu, \sigma^2/N)$,

$$E(\mathrm{SS}_{tr}) = k\sigma^2 + \sum n_i\mu_i^2 - \sigma^2 - N\mu^2 = (k-1)\sigma^2 + \sum n_i(\mu_i - \mu)^2,$$

using $\sum n_i\mu_i = N\mu$. This gives $E(\mathrm{MS}_{tr})$.

Step 4. The independence argument of the equal-replication case is unchanged. Under H_0, $\sqrt{n_i}(\bar{Y}_{i\cdot} - \mu)/\sigma$ are i.i.d. $N(0,1)$, and $\mathrm{SS}_{tr}/\sigma^2$ is the squared length of the projection of this vector onto the $(k-1)$-dimensional subspace orthogonal to $(\sqrt{n_1},\dots,\sqrt{n_k})$, hence $\chi^2(k-1)$. The F law follows. ■

Definition (Test of treatment effects (one-way, unequal replication))

$F = \mathrm{MS}_{tr}/\mathrm{MSE}$ with $\mathrm{MS}_{tr} = \mathrm{SS}_{tr}/(k-1)$, $\mathrm{MSE} = \mathrm{SSE}/(N-k)$, $N = \sum n_i$. Reject H_0 at level α if $f \geq F_\alpha(k-1, N-k)$; $P = P(F \geq f)$. The ANOVA table is as before with $\mathrm{SS}_{tr} = \sum_i n_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$.

Example 9-2 (Ratings on three review sites)

A film's ratings (out of 10) were sampled randomly: 4 from site 1, 5 from site 2, 6 from site 3. Test at level 5% whether the ratings differ by site.

	ratings	mean
Site 1	2, 3, 4, 5	3.5
Site 2	4, 5, 6, 4, 3	4.4
Site 3	6, 5, 7, 4, 6, 8	6.0

grand mean 4.8

Solution. $k = 3$, $n_1 = 4$, $n_2 = 5$, $n_3 = 6$; $H_0 : \alpha_1 = \alpha_2 = \alpha_3 = 0$.

$$\begin{aligned}\text{SST} &= (2-4.8)^2 + \cdots + (8-4.8)^2 = 36.4,\\ \text{SS}_{tr} &= 4(3.5-4.8)^2 + 5(4.4-4.8)^2 + 6(6.0-4.8)^2 = 16.2,\\ \text{SSE} &= (2-3.5)^2 + \cdots + (8-6.0)^2 = 20.2,\end{aligned}$$

with d.f. 14, 2, 12. $\text{MS}_{tr} = 8.1$, $\text{MSE} = 1.68$, $f = 4.82$, $P = P(F(2,12) \geq 4.82) = 0.0292$. The ratings differ significantly by site.

Source	SS	d.f.	MS	F	P
Treatment	16.2	2	8.1	4.82	0.0292
Residual	20.2	12	1.68		
Total	36.4	14			

9.3 Two-way analysis of variance without replication

Motivation. When two factors may influence the response, a two-way layout examines both at once; with one observation per combination we can still test each factor, provided the factors do not interact.

Two-way ANOVA studies the effects of two factors A (levels $A_1, \ldots, A_p$) and B (levels $B_1, \ldots, B_q$). Each combination of one level of A and one level of B is a treatment, so there are pq treatments. In a completely randomized design all pq runs are done in random order; fixing a level of A and running through the levels of B violates randomization. All factors here are fixed. With one observation y_{ij} per cell, define

$$\bar{y}_{i\cdot} = \frac{1}{q}\sum_{j=1}^{q} y_{ij}, \qquad \bar{y}_{\cdot j} = \frac{1}{p}\sum_{i=1}^{p} y_{ij}, \qquad \bar{y}_{\cdot\cdot} = \frac{1}{pq}\sum_{i=1}^{p}\sum_{j=1}^{q} y_{ij}.$$

Definition (Two-way ANOVA model (no replication))

$$Y_{ij} = \mu + \alpha_i + \beta_j + e_{ij}, \qquad e_{ij} \sim N(0,\sigma^2) \text{ independent}, \qquad \sum_{i=1}^{p}\alpha_i = 0, \quad \sum_{j=1}^{q}\beta_j = 0,$$

$i = 1, \ldots, p$, $j = 1, \ldots, q$; α_i is the effect of level A_i and β_j that of B_j.

The hypotheses of interest are

$$H_0 : \alpha_1 = \cdots = \alpha_p = 0 \ \text{ vs. } \ H_1 : \text{some } \alpha_i \neq 0; \qquad H_0 : \beta_1 = \cdots = \beta_q = 0 \ \text{ vs. } \ H_1 : \text{some } \beta_j \neq 0.$$

The total deviation decomposes as

$$y_{ij} - \bar{y}_{\cdot\cdot} = (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}) + (\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}) + (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}),$$

into the deviation due to A_i, that due to B_j, and the residual.

Proposition (Decomposition, two-way without replication)

$\mathrm{SST} = \mathrm{SS}_A + \mathrm{SS}_B + \mathrm{SSE}$, where

$$\mathrm{SST} = \sum_i \sum_j (y_{ij} - \bar{y}_{\cdot\cdot})^2 \quad (pq-1), \qquad \mathrm{SS}_A = q \sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 \quad (p-1),$$
$$\mathrm{SS}_B = p \sum_j (\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot})^2 \quad (q-1), \qquad \mathrm{SSE} = \sum_i \sum_j (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2 \quad ((p-1)(q-1)),$$

with degrees of freedom in parentheses.

Proof. [Supplement] Write $a_i = \bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$, $b_j = \bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot}$, $r_{ij} = y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}$. Then $\sum_i a_i = \sum_j b_j = 0$, and for each fixed i, $\sum_j r_{ij} = q\bar{y}_{i\cdot} - q\bar{y}_{i\cdot} - q\bar{y}_{\cdot\cdot} + q\bar{y}_{\cdot\cdot} = 0$; likewise $\sum_i r_{ij} = 0$ for each j.

Step 1. $\sum_i \sum_j a_i b_j = (\sum_i a_i)(\sum_j b_j) = 0$.

Step 2. $\sum_i \sum_j a_i r_{ij} = \sum_i a_i \sum_j r_{ij} = 0$ and $\sum_i \sum_j b_j r_{ij} = \sum_j b_j \sum_i r_{ij} = 0$.

Step 3. Squaring $y_{ij} - \bar{y}_{\cdot\cdot} = a_i + b_j + r_{ij}$ and summing, all cross terms vanish, leaving $q\sum a_i^2 + p\sum b_j^2 + \sum\sum r_{ij}^2$. The residuals satisfy $p+q-1$ independent linear constraints, so SSE has $pq-(p+q-1) = (p-1)(q-1)$ d.f.; and $(pq-1) = (p-1) + (q-1) + (p-1)(q-1)$. ■

Mean squares and F-statistics:

$$\mathrm{MS}_A = \frac{\mathrm{SS}_A}{p-1}, \quad \mathrm{MS}_B = \frac{\mathrm{SS}_B}{q-1}, \quad \mathrm{MSE} = \frac{\mathrm{SSE}}{(p-1)(q-1)}, \qquad F_A = \frac{\mathrm{MS}_A}{\mathrm{MSE}}, \quad F_B = \frac{\mathrm{MS}_B}{\mathrm{MSE}}.$$

Proposition (Expected mean squares, two-way without replication)

$E(\mathrm{MSE}) = \sigma^2$, $E(\mathrm{MS}_A) = \sigma^2 + \frac{q}{p-1}\sum \alpha_i^2$, $E(\mathrm{MS}_B) = \sigma^2 + \frac{p}{q-1}\sum \beta_j^2$. Under the respective null hypotheses $F_A \sim F(p-1, (p-1)(q-1))$ and $F_B \sim F(q-1, (p-1)(q-1))$.

Proof. [Supplement]

Step 1. $\bar{Y}_{i\cdot} = \mu + \alpha_i + \bar{e}_{i\cdot}$ (since $\sum_j \beta_j = 0$), and the $\bar{e}_{i\cdot}$ are independent $N(0, \sigma^2/q)$. As in the one-way proof, $E\sum_i (\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot})^2 = (p-1)\sigma^2/q + \sum \alpha_i^2$, so $E(\mathrm{SS}_A) = (p-1)\sigma^2 + q\sum \alpha_i^2$. Symmetrically for B.

Step 2. $\mathrm{SST} = \sum\sum (Y_{ij} - \bar{Y}_{\cdot\cdot})^2$ has expectation $(pq-1)\sigma^2 + \sum\sum (\alpha_i + \beta_j)^2 = (pq-1)\sigma^2 + q\sum \alpha_i^2 + p\sum \beta_j^2$ (the cross term vanishes by the constraints). Subtracting, $E(\mathrm{SSE}) = (pq - 1 - (p-1) - (q-1))\sigma^2 = (p-1)(q-1)\sigma^2$.

Step 3. The F laws follow from Cochran's theorem: $\mathrm{SS}_A, \mathrm{SS}_B, \mathrm{SSE}$ are quadratic forms in the errors with ranks $p-1, q-1, (p-1)(q-1)$ adding to $pq-1$, so under the null hypotheses they are independent $\sigma^2\chi^2$ variables with these d.f. ■

Definition (Tests of treatment effects (two-way, no replication))

(i) Factor A: $H_0 : \alpha_1 = \cdots = \alpha_p = 0$; $F_A = \mathrm{MS}_A/\mathrm{MSE}$; reject if $f_A \geq F_\alpha(p-1, (p-1)(q-1))$; $P_A = P(F_A \geq f_A)$.
(ii) Factor B: $H_0 : \beta_1 = \cdots = \beta_q = 0$; $F_B = \mathrm{MS}_B/\mathrm{MSE}$; reject if $f_B \geq F_\alpha(q-1, (p-1)(q-1))$; $P_B = P(F_B \geq f_B)$.

Source	SS	d.f.	MS	F	P
Factor A	SS_A	$p-1$	MS_A	$f_A = \mathrm{MS}_A/\mathrm{MSE}$	$P(F_A \geq f_A)$
Factor B	SS_B	$q-1$	MS_B	$f_B = \mathrm{MS}_B/\mathrm{MSE}$	$P(F_B \geq f_B)$
Residual	SSE	$(p-1)(q-1)$	MSE		
Total	SST	$pq-1$			

Example 9-3 (Diet and exercise)

Twenty men in their twenties of similar build were randomly assigned to one of 5 diets $(A_1, \ldots, A_5)$ combined with one of 4 exercises (B_1 running, B_2 weight training, B_3 swimming, B_4 yoga). Weight loss (kg) after 4 weeks:

	A_1	A_2	A_3	A_4	A_5	mean
B_1	7.9	7.2	5.1	5.8	6.8	6.56
B_2	7.5	6.6	4.8	5.6	6.5	6.20
B_3	6.9	6.4	4.4	5.4	6.4	5.90
B_4	6.5	6.2	4.1	4.5	5.8	5.42
mean	7.20	6.60	4.60	5.33	6.38	6.02

Test at level 5% whether diet and exercise affect weight loss.
Solution. $H_0 : \alpha_1 = \cdots = \alpha_5 = 0$ and $H_0 : \beta_1 = \cdots = \beta_4 = 0$.

$$\begin{aligned}
\mathrm{SS}_A &= 4\{(7.20-6.02)^2 + \cdots + (6.38-6.02)^2\} = 17.417,\\
\mathrm{SS}_B &= 5\{(6.56-6.02)^2 + \cdots + (5.42-6.02)^2\} = 3.492,\\
\mathrm{SST} &= (7.9-6.02)^2 + \cdots + (5.8-6.02)^2 = 21.232,\\
\mathrm{SSE} &= \mathrm{SST} - \mathrm{SS}_A - \mathrm{SS}_B = 0.323,
\end{aligned}$$

with d.f. $4, 3, 19, 12$. Then $\mathrm{MS}_A = 4.354$, $\mathrm{MS}_B = 1.164$, $\mathrm{MSE} = 0.027$, $f_A = 161.77$, $f_B = 43.24$, both with $P < 0.000001$. Both the diet and the exercise type have significant effects at level 5%.

Source	SS	d.f.	MS	F	P
Factor A	17.417	4	4.354	161.77	< 0.000001
Factor B	3.492	3	1.164	43.24	< 0.000001
Residual	0.323	12	0.027		
Total	21.232	19			

9.4 Two-way analysis of variance with replication

Motivation. Replicating each factor combination improves reliability and, crucially, lets us separate an *interaction* — an effect of A that changes with the level of B — from pure error.

Here factors A and B have p and q levels and each combination is replicated r times. The design must be completely randomized: all pqr runs in random order. (Randomizing only the order of the pq combinations and then doing the r runs of each consecutively is not a completely randomized design.)

The main advantage over the unreplicated design is that the effect arising from particular combinations of levels can be estimated separately. This is the **interaction**: the effect of A depends on

the level of B. Without replication the model has $E(Y_{ij}) = \mu + \alpha_i + \beta_j$, so the difference between two levels of A is the same at every level of B, e.g.

$$(\mu + \alpha_1 + \beta_j) - (\mu + \alpha_3 + \beta_j) = \alpha_1 - \alpha_3 \quad \text{for all } j.$$

Plotting the mean response against the levels of B, one line per level of A, gives parallel line segments: no interaction (Figure 1, left). If the order of the A-levels changes with B, the lines cross and A and B interact (Figure 1, right); then the unreplicated analysis is inappropriate and replication is needed.

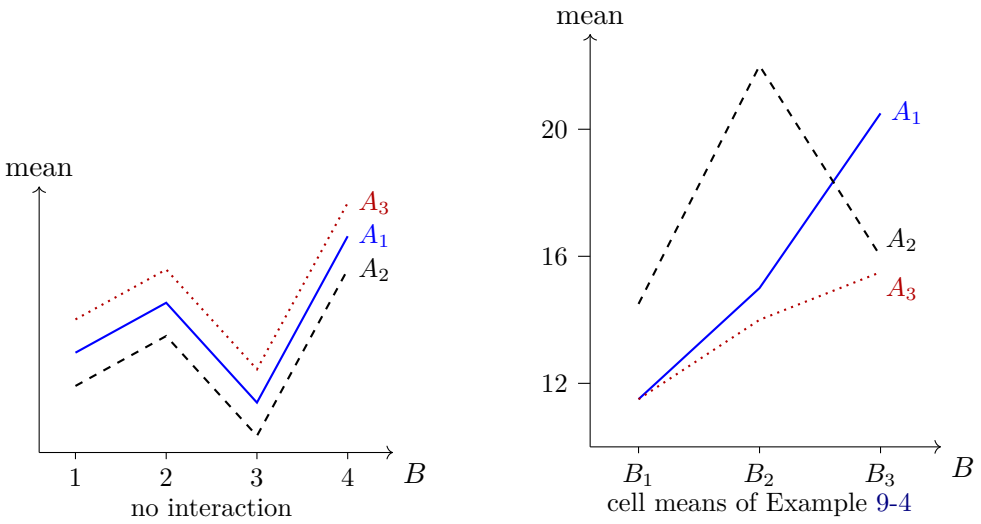


Figure 1: Interaction plots. Left: parallel profiles (no interaction). Right: crossing profiles, here the cell means $\bar{y}_{ij\cdot}$ of Example 9-4.

The data are y_{ijk} $(i = 1, \ldots, p;\ j = 1, \ldots, q;\ k = 1, \ldots, r)$ with means

$$\bar{y}_{ij\cdot} = \frac{1}{r}\sum_k y_{ijk}, \quad \bar{y}_{i\cdot\cdot} = \frac{1}{qr}\sum_j\sum_k y_{ijk}, \quad \bar{y}_{\cdot j\cdot} = \frac{1}{pr}\sum_i\sum_k y_{ijk}, \quad \bar{y}_{\cdots} = \frac{1}{pqr}\sum_i\sum_j\sum_k y_{ijk}.$$

Let γ_{ij} be the interaction effect of A_i and B_j.

Definition (Two-way ANOVA model (with replication))

$$Y_{ijk} = \mu + \alpha_i + \beta_j + \gamma_{ij} + e_{ijk}, \qquad e_{ijk} \sim N(0, \sigma^2) \text{ independent},$$

$$\sum_{i=1}^{p} \alpha_i = 0, \quad \sum_{j=1}^{q} \beta_j = 0, \quad \sum_{i=1}^{p} \gamma_{ij} = \sum_{j=1}^{q} \gamma_{ij} = 0,$$

$i = 1, \ldots, p,\ j = 1, \ldots, q,\ k = 1, \ldots, r.$

Tests of the main effects proceed as in §9.3; the new question is the interaction:

$$H_0 : \gamma_{ij} = 0 \ \ (\text{all } i, j) \qquad \text{vs.} \qquad H_1 : \text{at least one } \gamma_{ij} \neq 0.$$

The total deviation decomposes as

$$y_{ijk} - \bar{y}_{\cdots} = (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots}) + (\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots}) + (\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots}) + (y_{ijk} - \bar{y}_{ij\cdot}),$$

into deviations due to A_i, to B_j, to the interaction, and the residual.

Proposition (Decomposition, two-way with replication)

$\mathrm{SST} = \mathrm{SS}_A + \mathrm{SS}_B + \mathrm{SS}_{A\times B} + \mathrm{SSE}$, where

$$\begin{aligned} \mathrm{SST} &= \textstyle\sum_i \sum_j \sum_k (y_{ijk} - \bar{y}_{\cdots})^2 && (pqr-1),\\ \mathrm{SS}_A &= qr \textstyle\sum_i (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots})^2 && (p-1),\\ \mathrm{SS}_B &= pr \textstyle\sum_j (\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots})^2 && (q-1),\\ \mathrm{SS}_{A\times B} &= r \textstyle\sum_i \sum_j (\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots})^2 && ((p-1)(q-1)),\\ \mathrm{SSE} &= \textstyle\sum_i \sum_j \sum_k (y_{ijk} - \bar{y}_{ij\cdot})^2 && (pq(r-1)). \end{aligned}$$

Proof. [Supplement]

Step 1. Split $y_{ijk} - \bar{y}_{\cdots} = (\bar{y}_{ij\cdot} - \bar{y}_{\cdots}) + (y_{ijk} - \bar{y}_{ij\cdot})$. This is a one-way layout with the pq cells as treatments, each replicated r times, so by the one-way decomposition

$$\mathrm{SST} = r \textstyle\sum_i \sum_j (\bar{y}_{ij\cdot} - \bar{y}_{\cdots})^2 + \mathrm{SSE}.$$

Step 2. The cell means $\bar{y}_{ij\cdot}$ form a $p \times q$ table whose row means are $\bar{y}_{i\cdot\cdot}$, column means $\bar{y}_{\cdot j\cdot}$ and grand mean $\bar{y}_{\cdots}$. Applying the two-way decomposition of §9.3 to this table and multiplying by r,

$$r \textstyle\sum_i \sum_j (\bar{y}_{ij\cdot} - \bar{y}_{\cdots})^2 = \mathrm{SS}_A + \mathrm{SS}_B + \mathrm{SS}_{A\times B}.$$

Step 3. Combining gives the identity. D.f.: $pq(r-1)$ for SSE ($r-1$ per cell), and $pq - 1 = (p-1) + (q-1) + (p-1)(q-1)$ for the cell-mean part; the total is $pqr - 1$. ■

Mean squares and the interaction test statistic:

$$\mathrm{MS}_{A\times B} = \frac{\mathrm{SS}_{A\times B}}{(p-1)(q-1)}, \qquad \mathrm{MSE} = \frac{\mathrm{SSE}}{pq(r-1)}, \qquad F = \frac{\mathrm{MS}_{A\times B}}{\mathrm{MSE}}.$$

Proposition (Expected mean squares, two-way with replication)

$E(\mathrm{MSE}) = \sigma^2$, $E(\mathrm{MS}_A) = \sigma^2 + \dfrac{qr}{p-1}\sum \alpha_i^2$, $E(\mathrm{MS}_B) = \sigma^2 + \dfrac{pr}{q-1}\sum \beta_j^2$, and $E(\mathrm{MS}_{A\times B}) = \sigma^2 + \dfrac{r}{(p-1)(q-1)}\sum\sum \gamma_{ij}^2$. Under $H_0 : \gamma_{ij} = 0$, $F = \mathrm{MS}_{A\times B}/\mathrm{MSE} \sim F((p-1)(q-1), pq(r-1))$.

Proof. [Supplement]

Step 1. Within each cell, $Y_{ij1}, \ldots, Y_{ijr}$ is a sample from $N(\mu_{ij}, \sigma^2)$ with $\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$, so $E(\mathrm{SSE}) = pq(r-1)\sigma^2$ and $\mathrm{SSE}/\sigma^2 \sim \chi^2(pq(r-1))$, independent of all cell means (normal sampling theory, cells independent).

Step 2. The cell means $\bar{Y}_{ij\cdot} = \mu_{ij} + \bar{e}_{ij\cdot}$ are independent with variance σ^2/r, i.e. a two-way table without replication with error variance σ^2/r and "means" $\mu+\alpha_i+\beta_j+\gamma_{ij}$. By the computation in §9.3 (now with the extra term γ_{ij}, whose row and column sums vanish), $E(\mathrm{SS}_A) = (p-1)\sigma^2 + qr\sum \alpha_i^2$, $E(\mathrm{SS}_B) = (q-1)\sigma^2 + pr\sum \beta_j^2$ and

$$E(\mathrm{SS}_{A\times B}) = (p-1)(q-1)\sigma^2 + r \textstyle\sum_i \sum_j \gamma_{ij}^2.$$

Step 3. $\mathrm{SS}_{A\times B}$ is a function of the cell means, hence independent of SSE; under H_0 it is $\sigma^2\chi^2((p-1)(q-1))$ (Cochran's theorem applied to the cell-mean table, as in §9.3). The F law follows; the main-effect F-tests are justified the same way. ■

Definition (Test of interaction (two-way with replication))

For $H_0 : \gamma_{ij} = 0$ $(i = 1, \dots, p,\ j = 1, \dots, q)$ use $F = \mathrm{MS}_{A\times B}/\mathrm{MSE}$; reject at level α if $f \geq F_\alpha((p-1)(q-1), pq(r-1))$; $P = P(F \geq f)$. The main effects are tested by $F_A = \mathrm{MS}_A/\mathrm{MSE}$ and $F_B = \mathrm{MS}_B/\mathrm{MSE}$ with denominator d.f. $pq(r-1)$.

Source	SS	d.f.	MS	F	P
Factor A	SS_A	$p-1$	MS_A	$f_A = \mathrm{MS}_A/\mathrm{MSE}$	$P(F_A \geq f_A)$
Factor B	SS_B	$q-1$	MS_B	$f_B = \mathrm{MS}_B/\mathrm{MSE}$	$P(F_B \geq f_B)$
Interaction	$\mathrm{SS}_{A\times B}$	$(p-1)(q-1)$	$\mathrm{MS}_{A\times B}$	$f_{A\times B} = \mathrm{MS}_{A\times B}/\mathrm{MSE}$	$P(F_{A\times B} \geq f_{A\times B})$
Residual	SSE	$pq(r-1)$	MSE		
Total	SST	$pqr-1$			

Example 9-4 (Machines and operators)

Three machines (A_1, A_2, A_3) and three operators (B_1, B_2, B_3) were compared by measuring productivity twice for each combination. Test the main effects and the interaction at level 5%.

	B_1	B_2	B_3	mean
A_1	9, 14	14, 16	19, 22	15.67
A_2	13, 16	18, 26	14, 18	17.5
A_3	11, 12	11, 17	15, 16	13.67
mean	12.5	17	17.33	15.61

Solution. Hypotheses: $H_0 : \alpha_1 = \alpha_2 = \alpha_3 = 0$; $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$; $H_0 : \gamma_{ij} = 0$ $(i, j = 1, 2, 3)$.

$$\begin{aligned}
\mathrm{SS}_A &= 3 \times 2\{(15.67 - 15.61)^2 + \cdots + (13.67 - 15.61)^2\} = 44.11,\\
\mathrm{SS}_B &= 3 \times 2\{(12.5 - 15.61)^2 + \cdots + (17.33 - 15.61)^2\} = 87.44,\\
\mathrm{SS}_{A\times B} &= 2\{(11.5 - 15.67 - 12.5 + 15.61)^2 + \cdots\} = 74.22,\\
\mathrm{SST} &= (9 - 15.61)^2 + \cdots + (16 - 15.61)^2 = 288.28,\\
\mathrm{SSE} &= \mathrm{SST} - \mathrm{SS}_A - \mathrm{SS}_B - \mathrm{SS}_{A\times B} = 82.51,
\end{aligned}$$

with d.f. $2, 2, 4, 17, 9$. Hence $\mathrm{MS}_A = 22.06$, $\mathrm{MS}_B = 43.72$, $\mathrm{MS}_{A\times B} = 18.56$, $\mathrm{MSE} = 9.17$ and $f_A = 2.41$, $f_B = 4.77$, $f_{A\times B} = 2.02$, with P-values 0.146, 0.039, 0.174. Operators differ significantly in productivity at level 5%, while the machine type and the interaction are not significant.

Source	SS	d.f.	MS	F	P
Factor A	44.11	2	22.06	2.41	0.146
Factor B	87.44	2	43.72	4.77	0.039
Interaction	74.22	4	18.56	2.02	0.174
Residual	82.51	9	9.17		
Total	288.28	17			

Model checking and beyond

As in regression, the error assumptions should be checked before testing, using a residual plot of studentized residuals against the estimated mean responses. For Example 9-4 this plot

shows no problems, so the tests are meaningful. Once a factor is found significant, one would go on to estimate the level means or compare them; this lies beyond the scope of these notes.